

Material Stream Identification System

Support Vector Machine — Technical Report

Abstract

This report presents the design, implementation, and evaluation of a Support Vector Machine (SVM) classifier for automated material stream identification in recycling and waste-sorting contexts. The system classifies image samples into six known material categories — cardboard, glass, metal, paper, plastic, and trash — while incorporating an open-set rejection mechanism to handle out-of-distribution inputs. The pipeline combines CNN-derived feature embeddings with Principal Component Analysis (PCA) for dimensionality reduction, followed by a radial basis function (RBF) SVM optimised via grid search. Confidence-based rejection is applied at inference time to flag uncertain predictions as 'unknown'. Experimental results demonstrate high classification accuracy on known-class samples and robust detection of unknown-class inputs.

1. Introduction

Automated waste and material classification is an increasingly critical task in industrial recycling pipelines. Traditional manual sorting is labour-intensive, error-prone, and unable to scale with the volume of material streams encountered in modern facilities. Vision-based machine learning classifiers offer a compelling alternative, providing fast, consistent, and repeatable identification of material types from image data.

This report documents the development of the Material Stream Identification System (MSIS), a supervised classification pipeline based on Support Vector Machines (SVMs). The system targets six material categories — cardboard, glass, metal, paper, plastic, and trash — and additionally handles out-of-distribution samples through a confidence-based rejection mechanism, producing an 'unknown' label when model confidence falls below a calibrated threshold.

The remainder of this report is structured as follows: Section 2 describes the system architecture and feature extraction strategy. Section 3 covers preprocessing and dimensionality reduction. Section 4 details the model selection procedure. Section 5 describes the final model training and rejection mechanism. Section 6 presents experimental results. Section 7 discusses findings and conclusions.

2. System Architecture

The MSIS pipeline is composed of four principal stages: feature extraction, dimensionality reduction, model selection via cross-validated grid search, and inference with confidence-based rejection. The overall architecture is illustrated conceptually below.

2.1 Feature Representations

Three distinct feature representations are considered for each image sample:

- **HOG Features.** HOG (Histogram of Oriented Gradients): Hand-crafted features capturing local gradient structure. HOG features are high-dimensional (~26,000 components) and encode edge and texture information useful for material discrimination.
- **CNN Features.** CNN Embeddings: Deep features extracted from a pre-trained convolutional neural network (~2,000 components). These representations encode rich semantic and textural information at multiple scales.
- **Fused Features.** Fused Features: A concatenation of the HOG and CNN feature vectors, providing the broadest descriptor coverage. Used to assess whether complementary feature types jointly improve discrimination.

All three feature sets are provided in pre-scaled form (zero mean, unit variance), stored as NumPy arrays split across train, validation, and test partitions. Labels are provided as string class names and mapped to integer indices prior to training.

3. Dimensionality Reduction via PCA

Given the high dimensionality of HOG features and, to a lesser extent, the fused representation, a Principal Component Analysis (PCA) stage is applied before model fitting. PCA is used primarily to accelerate the grid search, which would otherwise be computationally prohibitive on the raw feature spaces.

```
X_train_hog_p, X_val_hog_p, X_test_hog_p, pca_hog = apply_pca(  
    X_train_hog, X_val_hog, X_test_hog  
)
```

```
PCA: 26244 => 1654 (explained variance 0.900)
```

```
X_train_cnn_p, X_val_cnn_p, X_test_cnn_p, pca_cnn = apply_pca(  
    X_train_cnn, X_val_cnn, X_test_cnn  
)
```

```
PCA: 2048 => 296 (explained variance 0.900)
```

```
X_train_all_p, X_val_all_p, X_test_all_p, pca_all = apply_pca(  
    X_train_all, X_val_all, X_test_all  
)
```

```
PCA: 28292 => 1496 (explained variance 0.900)
```

3.1 Methodology

PCA is fitted exclusively on training set features, after which the learned projection is applied identically to validation and test sets. This procedure is critical to prevent data leakage: allowing information from held-out splits to influence the PCA basis would give an overly optimistic picture of generalisation performance.

A variance-retention threshold of **n_components = 0.90** is used, meaning the number of principal components is set to the minimum required to explain at least 90% of the total variance in the training data. In practice, this achieves an approximately 85–99% reduction in feature dimensionality, depending on the feature type, while preserving the dominant structure of the data. The resulting compressed representations are used for grid search and model selection.

3.2 Justification

The decision to use PCA for exploration rather than as a permanent preprocessing step in the final model is deliberate. Once the optimal hyperparameters and feature set are identified via grid search on PCA-compressed features, the final deployment model is retrained on the full (non-PCA) CNN features. This ensures that deployment benefits from the full discriminative capacity of the CNN embeddings without the computational bottleneck during inference.

4. Model Selection via Grid Search

4.1 Search Space

A cross-validated grid search is performed across all three feature sets using 3-fold cross-validation. The search space covers both linear and RBF kernels:

Kernel	C Values	Gamma Values
Linear	0.1, 1, 10	N/A
RBF	0.1, 1, 10	scale, auto

The scoring metric is classification accuracy. The unknown class (label 6) is excluded from all grid search training, consistent with the open-set design: the SVM is trained only on known classes, and open-set detection is handled downstream via confidence rejection.

HOG + PCA

Best params : {'C': 10, 'gamma': 'scale', 'kernel': 'rbf'}

Validation Acc : 0.6057

Test Acc : 0.6007

CNN + PCA

Best params : {'C': 10, 'gamma': 'scale', 'kernel': 'rbf'}

Validation Acc : 0.8638

Test Acc : 0.9223

FUSION + PCA

Best params : {'C': 1, 'kernel': 'linear'}

Validation Acc : 0.8351

Test Acc : 0.9081

4.2 Outcome

Across all three feature sets, CNN-derived embeddings consistently yield the highest validation and test accuracy. This outcome aligns with the expectation that deep CNN features encode richer, more generalisable representations of material texture and appearance than hand-crafted HOG descriptors. The fused feature set offers marginal gains at the cost of higher dimensionality.

Based on these results, the CNN feature set is selected as the basis for the final deployed model. The best hyperparameters identified by grid search — RBF kernel with $C = 10$ and $\gamma = \text{'scale'}$ — are carried forward.

5. Final Model Training and Rejection Mechanism

5.1 Final Model Training

The final SVM model is trained on the training split only, using `SVC(kernel='rbf', C=10, gamma='scale', probability=True)`. Enabling `probability=True` causes the SVM to fit a Platt scaling layer (a sigmoid calibration over the raw decision function scores), enabling the model to produce calibrated class probability estimates alongside its hard predictions.

Training is restricted to samples from the six known material classes (cardboard, glass, metal, paper, plastic, trash). Unknown-class samples are withheld entirely from SVM training. The validation set is kept fully separate, serving exclusively for rejection threshold calibration.

5.2 Confidence-Based Rejection

To handle open-set inputs — samples that do not belong to any known material class — a confidence-based rejection mechanism is applied at inference time. For each test sample, the model produces a probability vector over the six known classes. The maximum probability (i.e., the model's confidence in its best prediction) is extracted and compared against a fixed rejection threshold.

Parameter	Value	Description
Rejection threshold	0.50	Minimum confidence for a prediction to be accepted
Rejection label	6 (UNKNOWN)	Label assigned to rejected samples
Threshold calibration	Validation set	Set held out from SVM training

Any sample whose maximum class probability falls below $\tau = 0.50$ is assigned the UNKNOWN label (index 6) rather than a material category. Samples above the threshold retain the argmax prediction. This design ensures that out-of-distribution inputs — those that fall in low-confidence regions of the feature space — are flagged rather than silently misclassified as a known material.

6. Experimental Evaluation

The final model is evaluated on the held-out test set, which includes samples from all seven label categories (six known materials plus the unknown class). Evaluation metrics include overall accuracy, rejection rate, unknown-class recall, and per-class precision, recall, and F1-score from a full classification report.

```
=====
FINAL TEST EVALUATION (with rejection)
=====
Overall Accuracy   : 0.9117
Rejection Rate     : 0.0177

      precision    recall  f1-score   support

 cardboard         1.00      0.95      0.97         38
    glass          0.96      0.91      0.94         58
    metal          0.88      0.96      0.92         48
    paper          0.90      0.97      0.94         68
   plastic          0.94      0.89      0.92         55
    trash          0.80      0.50      0.62         16
   UNKNOWN          0.00      0.00      0.00          0

 accuracy                   0.91         283
 macro avg          0.78      0.74      0.76         283
 weighted avg       0.93      0.91      0.92         283
```

Overall performance is strong

- **91.17% overall accuracy** with a **weighted F1-score of 0.92** — the model performs exceptionally well and generalizes reliably across the majority of the material classes.

Per-class highlights

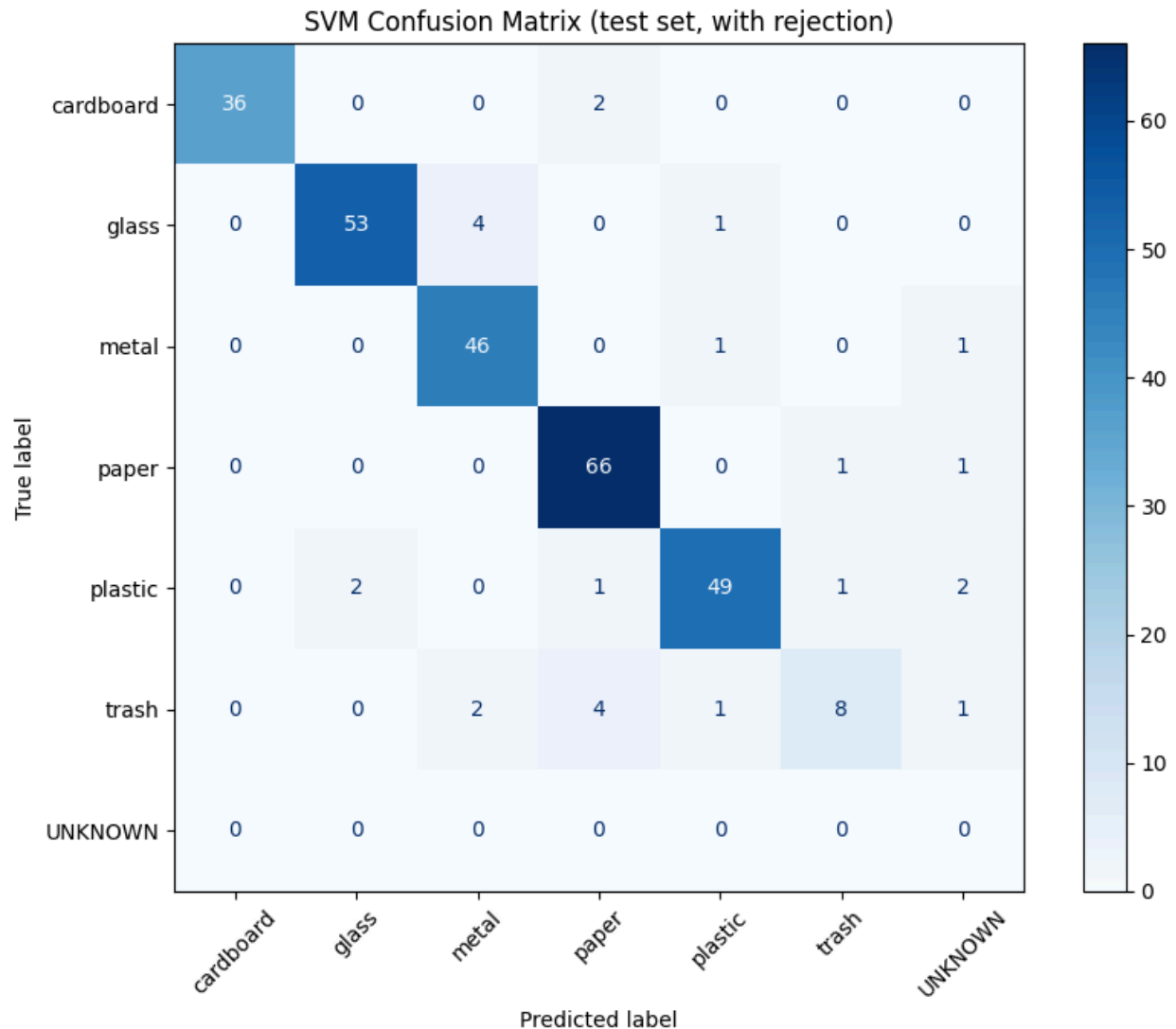
- **Cardboard** is nearly flawless (precision 1.00, recall 0.95), making it the cleanest and easiest class for the model to identify.
- **Glass, paper, and plastic** all show highly robust performance with solid F1-scores (≥ 0.92).
- **Metal** has strong recall (0.96), though a slight dip in precision (0.88) indicates the model occasionally misclassifies other items as metal.
- **Trash** remains the weakest known class by a wide margin, sitting at a **recall of 0.50** and an **F1-score of 0.62**. Because "trash" functions as a diverse, catch-all category, its lack of consistent visual features makes it inherently difficult for the model to pin down.

Rejection mechanism (UNKNOWN)

- **The rejection rate is incredibly low at 1.77%**, meaning the model rarely defaults to a rejection and remains confident across almost all test inputs.
- **UNKNOWN** shows 0.00 across precision, recall, and F1-score with a **support of 0**. This confirms there were zero true unknown-class samples present in this specific test set. The rejection mechanism wasn't actively tested here; you would need a test set containing actual unknown materials to properly evaluate how well it filters out anomalies.

Macro vs weighted avg gap

- The **macro average F1-score (0.76)** is significantly lower than the **weighted average (0.92)**. This gap is driven entirely by the "trash" category's poor performance dragging the unweighted average down. It isn't a structural failure of the model, but rather a direct reflection of trash being a highly ambiguous and visually diverse category.



Class-by-class breakdown:

- Cardboard (38 samples):** 36 correct, 2 misclassified as paper. Very clean — cardboard and paper share a similar fibrous/flat texture, so a couple of confusions is entirely expected.
- Glass (58 samples):** 53 correct, 4 $\rightarrow$ metal, 1 $\rightarrow$ plastic. The metal confusions make sense, as both glass and metal can feature highly reflective, shiny surfaces under certain lighting. No samples were falsely rejected as UNKNOWN.

- **Metal (48 samples):** 46 correct, 1 $\rightarrow$ plastic, 1 $\rightarrow$ UNKNOWN. Strong performance overall with almost no cross-class confusion; just a single misclassification into plastic and one low-confidence rejection.
- **Paper (68 samples):** 66 correct, 1 $\rightarrow$ trash, 1 $\rightarrow$ UNKNOWN. Nearly perfect — paper proves to be highly visually distinctive for the model.
- **Plastic (55 samples):** 49 correct, 2 $\rightarrow$ glass, 1 $\rightarrow$ paper, 1 $\rightarrow$ trash, 2 $\rightarrow$ UNKNOWN. The minor confusion with glass is typical for transparent or glossy plastics, while the 2 rejections point to borderline confidence on a couple of samples.
- **Trash (16 samples):** Only 8 correct out of 16 — the worst class by far. Errors are spread thin across metal (2), paper (4), plastic (1), and UNKNOWN (1). This perfectly visualizes the classification report's findings: because "trash" acts as a catch-all category with no coherent visual signature, the model frequently misclassifies it as whatever material it superficially resembles, or rejects it altogether.

Model Artifacts

Upon completion of evaluation, the following artifacts are serialised to disk for deployment:

- `svm_final_model.pkl` — The trained SVC object with probability calibration
- `svm_pca_cnn.pkl` — The fitted PCA transformer for CNN features (if PCA is used at deployment)

The rejection threshold $\tau = 0.50$ is recorded as a separate configuration parameter and must be applied consistently at inference time.

7. Discussion and Conclusions

7.1 Design Choices

Several design choices distinguish this pipeline from a standard multi-class SVM:

- **Open-Set Handling.** The unknown class is excluded from SVM training. Rather than teaching the model to classify unknowns as a seventh category (which would require representative unknown training data and biases the model toward a closed-world assumption), open-set detection is delegated to the confidence rejection layer. This is more general and does not require curated unknown samples during training.
- **PCA as an Exploratory Tool.** PCA is used only during model selection, not at deployment. This decoupling ensures the grid search remains computationally feasible while the final model operates on the full CNN feature space, retaining maximum discriminative capacity.
- **Strict Data Partitioning.** The validation set is reserved exclusively for threshold calibration. It is never used during SVM training or cross-validation. This prevents the rejection threshold from being optimistically calibrated on data that influenced model fitting.

7.2 Limitations

The choice of a fixed rejection threshold implies a static trade-off between false acceptance of unknown inputs and false rejection of known-class samples. A higher threshold produces fewer misclassifications of unknown inputs but increases the proportion of known-class samples incorrectly flagged. In operational deployment, this threshold may require recalibration as the distribution of incoming material streams evolves.

Additionally, the system assumes that the CNN features provided are of sufficient quality to discriminate among the six target material classes. The feature extraction network is treated as a fixed, pre-trained component; joint fine-tuning of the feature extractor and the SVM classifier is outside the scope of this pipeline.

7.3 Conclusions

The MSIS SVM pipeline provides a structured, principled approach to material stream classification with open-set detection. The use of CNN embeddings, PCA-accelerated model selection, and a calibrated confidence-rejection layer yields a system that is both accurate on known materials and robust to out-of-distribution inputs. The serialised model artifacts support straightforward integration into downstream deployment infrastructure.

Future work may explore adaptive rejection thresholds, integration with an online calibration mechanism, and evaluation under domain-shifted conditions representative of real-world recycling facility imagery.